



Python for Fixed Income

COURSE 4 · MACHINE LEARNING

ROC Curves and Choosing a Threshold

WEEK 2 · DAY 3

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Where day 7 left off

Day 7: threshold 0.5

324 of 1,327

defaults caught, 6,000 validation rows

Recall 0.2442, precision 0.7248. Nothing about 0.5 says it is the place to stop.

Real data

UCI card file: accounts in Taiwan, 2005. Not US credit today.

People's attributes

SEX, EDUCATION, MARRIAGE, AGE: never features.

60 / 20 / 20

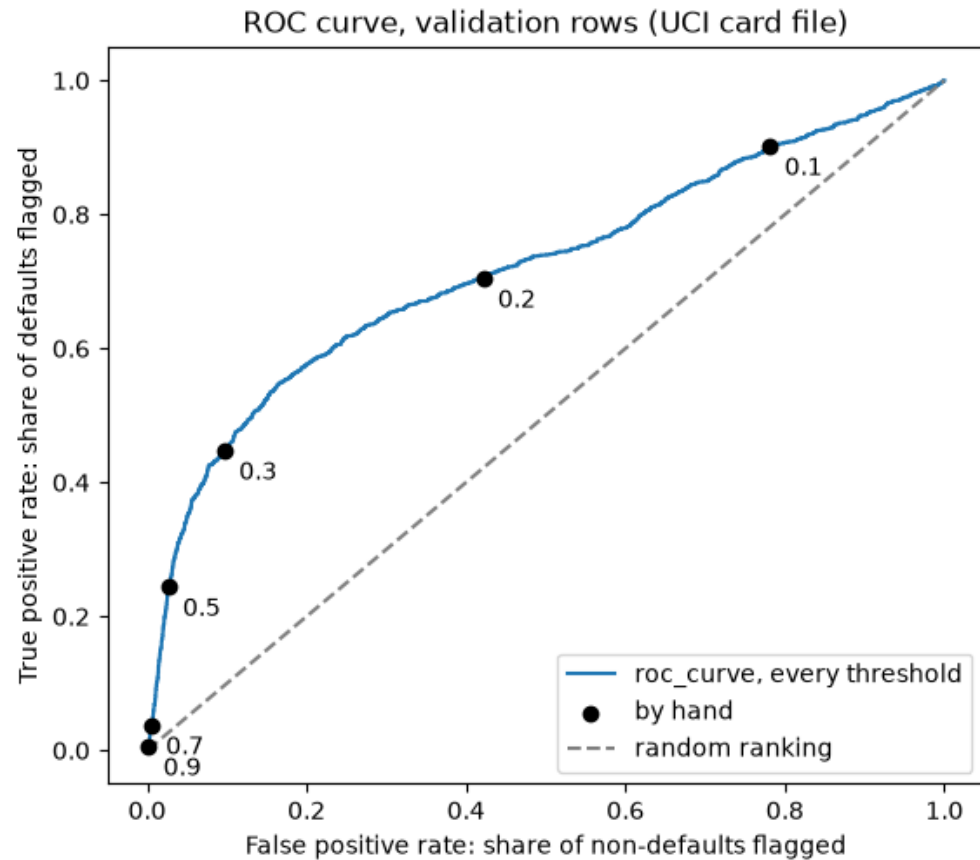
Training fits, validation chooses, test reports once (day 10).

Names

Model probability of default in this file; flagged at threshold t .

Every threshold and target today is a teaching convention, not what a lender does or should do.

Every threshold at once: the ROC curve



Rate	Formula	Out of
True positive (recall)	$TP / (TP + FN)$	1,327 defaults
False positive	$FP / (FP + TN)$	4,673 others

Threshold	Flagged	TPR	FPR
0.5	447	0.2442	0.0263
0.3	1,047	0.4476	0.0969
0.2	2,905	0.7046	0.4216
0.1	4,848	0.9020	0.7813

Up the curve is a lower threshold. The diagonal is a score with no information.

AUC as a ranking measure

0.7190

ROC AUC, validation rows

6,201,071

default and non-default pairs
compared

Pick one account that defaulted and one that did not: AUC is the chance the model ranks the defaulted one higher (ties count half). Here, 71.90 percent of pairs.

What AUC does not say

Not an accuracy, not a default rate

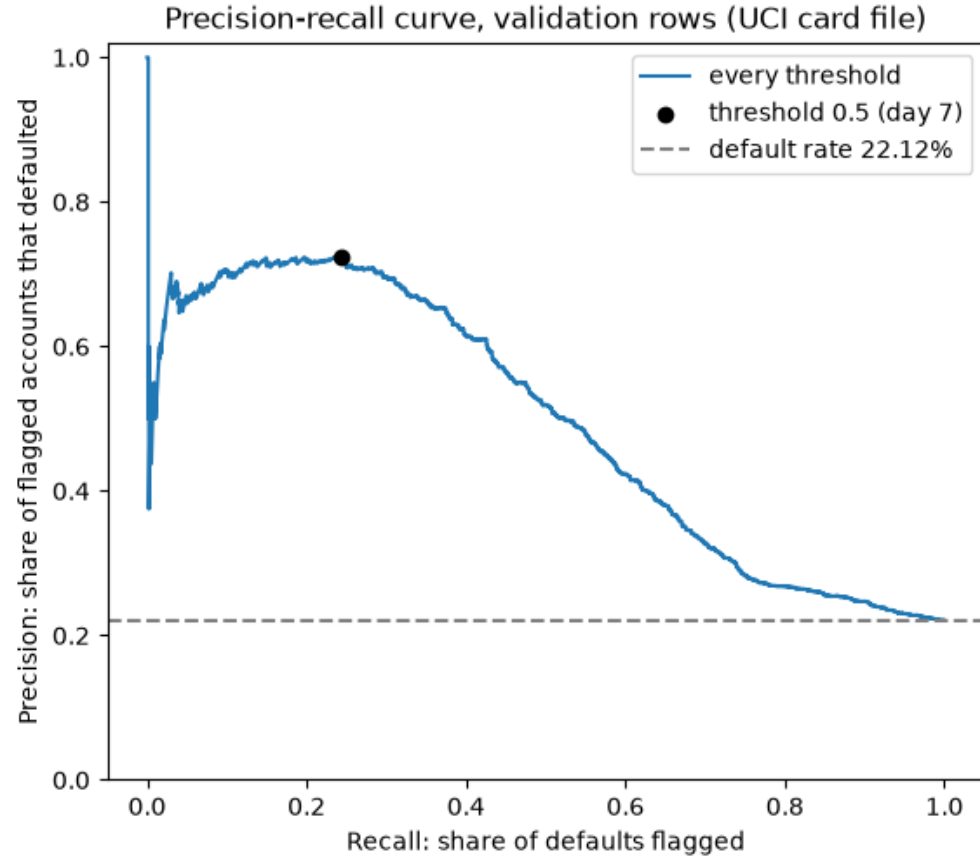
Nothing about any one threshold

Not whether a 0.3 means three in ten

Squared probabilities: still 0.7190

On probabilities, never on 0/1 flags: the flags at 0.5 give 0.6089, a different number.

The precision-recall curve



0.4946

average precision

22.12%

the floor: the default rate

Precision divides by the flagged accounts: of those flagged, how many defaulted? The false positive rate divides by every non-default.

Average precision is a step sum: each rise in recall times the precision there. Not a trapezoid.

Choosing a threshold on validation rows

The course's rule: `mlkit.threshold_for_recall`

If the desk asked for recall of 70 percent: the highest threshold with validation recall at least 0.70.

Rule	Threshold	Flagged	Recall	Precision
Day 7: 0.5	0.5000	447	0.2442	0.7248
Recall at least 0.70	0.2025	2,843	0.7001	0.3268
Youden's J	0.2475	1,507	0.5494	0.4837

Youden's J: where TPR minus FPR is largest (0.3829). Accuracy of flagging nothing: 0.7788; never used to choose.

A teaching convention, not a lending standard. No threshold here is what a lender does or should do.

Why the test rows are not used to choose

The validation rows cut in half 20 times: choose the threshold on one half, measure recall on both.

0.7003 to 0.7014

recall on the half that chose: never below 0.70

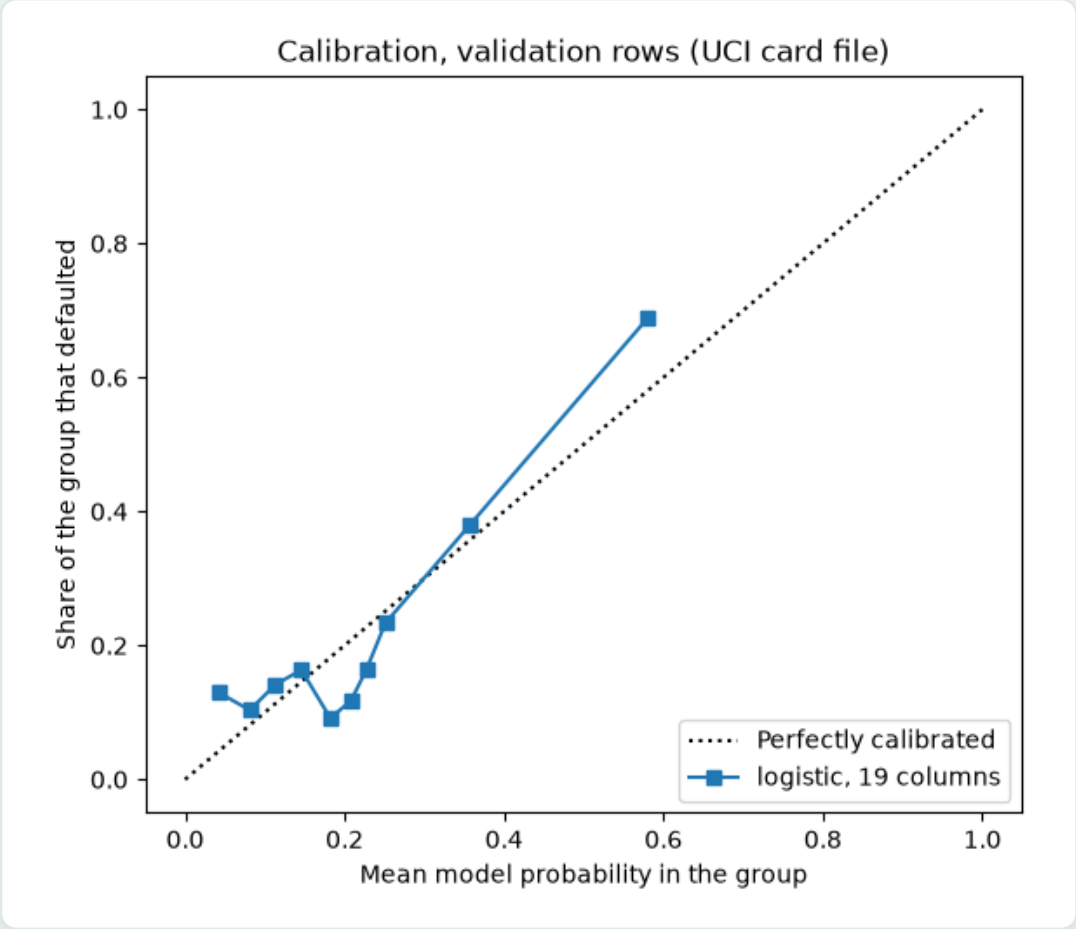
0.6395 to 0.7376

recall on the other half: below 0.70 in 15 of 20

The threshold itself moved from 0.1941 to 0.2160 with the rows. Rows that choose a setting always look good on it.

Chosen on the test rows, the reported recall would be 0.70 by construction. Day 10 opens them once.

Calibration: does a 0.3 mean three in ten?



Group	Mean probability	Defaulted
Lowest	0.0416	13.00%
Nearest 0.3	0.2512	23.33%
Top	0.5800	68.83%

Ten groups of about 600 accounts, sorted by probability. On the diagonal, a probability means what it says.

It ranks better than chance; its probabilities are not uniformly reliable as numbers. Recalibration: Course 5.

Excel side by side

Excel side by side (tested in Excel)

=COUNTIFS(A2:A6001,1,B2:B6001,">="&D2) (row 2)

=F2/COUNTIF(A2:A6001,1)

=SORT(UNIQUE(\$B\$2:\$B\$6001),,-1) (M3)

=COUNTIFS(\$A\$2:\$A\$6001,1,\$B\$2:\$B\$6001,">="&M3#)/COUNTIF(\$A\$2:\$A\$6001,1) (O3)

=SUMPRODUCT((N3:N5884-N2:N5883),(O3:O5884+O2:O5883)/2)

=MAXIFS(M3#,O3#,">=0.7")

=INDEX(M3#,MATCH(MAX(O3#-N3#),O3#-N3#,0))

=SUMPRODUCT((O3:O5884-O2:O5883),P3:P5884)

TP 324 at 0.5

recall 0.2442

5,882 thresholds

TPR, spilled

AUC 0.7190

rule 0.2025

Youden 0.2475

AP 0.4946

Excel has no ROC or AUC function: the probabilities come from Python; every step after them is a formula.

Save the day: the history of the choices

```
git log --oneline -- results.csv
Experiment log: Youden's J threshold beside it, for comparison
Experiment log: the threshold for recall 0.70, chosen on validation rows
Experiment log: the logistic model at day 7's threshold 0.5
```

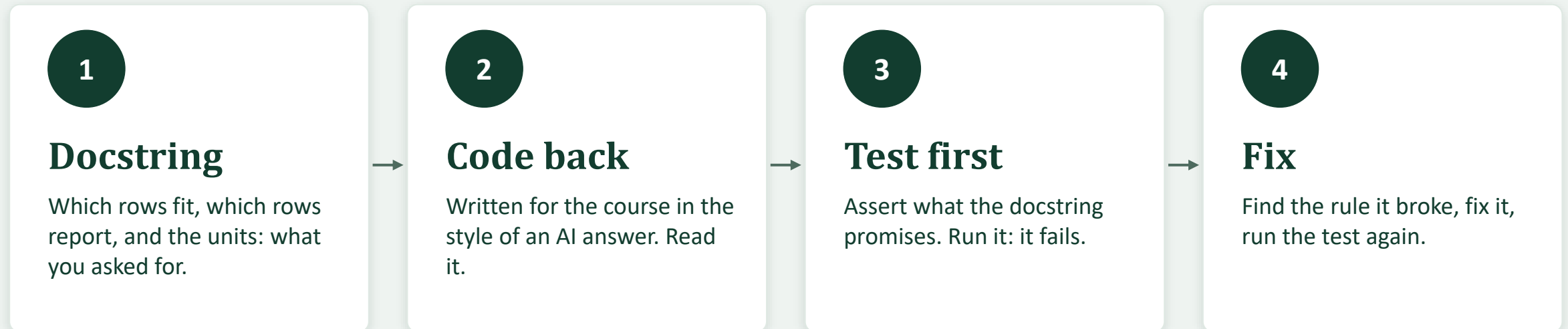
```
model,rows,roc_auc,average_precision,threshold,accuracy,precision,recall,f1
logistic_scaled at fixed 0.5,validation,0.7190,0.4946,0.5000,0.8123,0.7248,0.2442,0.3653
logistic_scaled at recall at least 0.70,validation,0.7190,0.4946,0.2025,0.6147,0.3268,0.7001,0.4456
logistic_scaled at Youden J,validation,0.7190,0.4946,0.2475,0.7707,0.4837,0.5494,0.5145
```

One row per choice, one commit per row. The two dashes keep only the commits that changed results.csv.

Why the threshold is what it is: the log shows the order of the choices and the commit behind each.

The AI check

The docstring is the prompt. The code that comes back runs, and has one modelling bug.



Today: the threshold is chosen on the test rows, with a comment that says the model never saw them.

Code that runs is not the same as code that is right.

What you can do now

- The ROC curve: every threshold at once, each point a confusion matrix
- AUC 0.7190: pairs ranked the right way, nothing about one threshold
- Average precision 0.4946: a step sum over the precision-recall curve
- Recall at least 0.70 on validation rows: threshold 0.2025, a convention
- The rows that choose cannot report; calibration checks the numbers

Week 2, Day 4 Decision trees: a model made of questions, on the same validation rows